

A δ^2 – L^3 Sensitivity Law for Finite-Window Weil Positivity, with a Function-Field Control of the Connes–van Suijlekom Pipeline

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October 2026 — Technical Note

Concept DOI (all versions): doi:10.5281/zenodo.21109955

This version (v10): doi:10.5281/zenodo.23224986

Previous version (v9): doi:10.5281/zenodo.23212998

Abstract

The Connes–van Suijlekom and Connes–Consani–Moscovici programs reduce the Riemann hypothesis to spectral properties of a quadratic form restricted to test functions supported in a finite window. We ask how sensitive that form is to a zero moved off the critical line, and we calibrate the answer in a setting where the truth is known: curves over finite fields, where the window form is exactly the Frobenius Gram matrix of Hallouin–Perret and the Riemann hypothesis is Weil’s theorem. Our main result is a sensitivity law. In the function field we prove that moving a reciprocal real pair $\{\rho, \rho^{-1}\}$ off the critical circle gives $\lambda_{\min}(T_R) = -\binom{R+2}{3}a^2 + O(a^4)$ with $a = \log \rho$, the coefficient being twice the centred second moment of the window. On the zeta side, in the even sector, we prove a bound $\lambda_{\min}(Q_\delta) \geq \lambda^*(L) - c(L)\delta^2 - O(\delta^4)$ with $c(L) = 8L^3(\frac{1}{3} + \frac{1}{\sqrt{5}})$, stated *relative* to a lower bound $\lambda^*(L)$ for the unperturbed form and unconditional where that positivity is certified ($L \leq 0.8$), and a sharper conditional bound with the *exact* constant $4K(L, \gamma)$, where $K(L, \gamma) = \int_{-L}^L u^2 \sin^2(\gamma u) du$ is attained by $f \propto u \sin(\gamma u)$; numerically the re-optimised minimiser reaches up to 96% of it. Under an explicit dictionary the two laws are the same expression, (window length)³/6. We also run the full Connes–van Suijlekom minimiser pipeline on deliberately planted off-circle spectra, which separates the steps that carry arithmetic content from those that do not. A final section identifies the mechanism behind the function-field coefficient — it is the squared norm of a derivative-evaluation functional compressed to the kernel of the unperturbed form, which we prove for a deformation family containing the one above as its degenerate member — and reports, numerically and separately, how much of the measured ζ -side response that same overlap accounts for, together with a parity hypothesis that the data refutes. **Nothing here proves or advances the Riemann hypothesis for ζ .**

Non-claims

Stated first, not in an appendix.

1. **No proof or advance of RH for ζ .** Part I runs where the Riemann hypothesis is a theorem (Weil, curves over finite fields); Part II uses fabricated spectra.
2. **No new spectral architecture.** The window form, its Toeplitz structure and its Carathéodory–Fejér theory are due to Hallouin–Perret [8] and Connes–van Suijlekom [5]; the finite-window Weil form itself goes back to Yoshida [10] and Bombieri [1].

3. **Planted spectra are fabricated diagnostics**, not L -functions of anything.
4. **Numerical detections are tier T2 unless certified**, and *non-detections certify nothing*: a finite basis that finds no negative direction is not evidence of positivity. This asymmetry is the subject of §6.
5. $\lambda^*(0.8) \leq 2.2702 \times 10^{-17}$ **is a variational upper bound** from a finite basis, not a certified enclosure. We prove no rigorous lower bound *of our own*; Zhu [11] certifies one, and Theorem 7 uses it to convert its relative estimate into an unconditional lower bound for $L \leq 0.8$. Theorem 8 does *not* use it; it rests on its own hypothesis.
6. **§5 reproduces published computations** (Groskin [6]) at smaller scale.
7. **Block F is claimed only for the function field**. On ζ windows the indefiniteness of A and M depends on where $-\log \pi$ is placed; see §7 and Retraction 15.
8. **Not attempted**: Davenport–Heilbronn; any Lean formalisation.

Epistemic tiers

Tier	Meaning	Discharged by
T1	exact or proved	a proof, or exact arithmetic over $\mathbb{Z}$ or $\mathbb{Q}$
T2	numerical	stated working precision + a convergence or noise-floor check
T3	interpretation	argument from T1/T2 evidence, labelled as such

Every numbered statement below carries a tier, the script that produced it and the data file that records it. All numerics use `mpmath` at the stated precision; `float64` is used only as a deliberate contrast (§11).

1 The pipeline and the question

Let $X/\mathbb{F}_q$ be a smooth projective absolutely irreducible curve of genus g , and let $N_n = \#X(\mathbb{F}_{q^n})$. Hallouin and Perret [8, Prop. 5] show that the normalised classes γ^k of the iterated Frobenius graphs have Gram matrix

$$\text{Gram}(\gamma^0, \dots, \gamma^R) = (t(|i - j|))_{0 \leq i, j \leq R}, \quad t(0) = 2g, \quad t(n) = \frac{(q^n + 1) - N_n}{q^{n/2}}, \quad (1)$$

a symmetric Toeplitz matrix, which we write T_R . We adopt (1) as the definition of the window form; it is *their* matrix, entry for entry.

The Connes–van Suijlekom pipeline [5] passes through four steps: (i) reality of the zeros of the ground-state polynomial; (ii) the sign of $\lambda_{\min}$; (iii) simplicity and evenness of the ground state; (iv) convergence of the normalised minimiser. The question of this note is *quantitative*: how large a negative eigenvalue does a zero moved a distance δ off the critical line actually produce, and how large a window is needed to see it?

What is claimed as new. Only three things.

- (N1) The δ^2 – L^3 sensitivity law: Theorem 6 (function field), Theorems 7 and 8 (ζ), the measured near-saturation, and the shared (window length)³/6 form.
- (N2) The planted-spectrum control of the CvS minimiser pipeline in the function field (§3).

(N3) The projected-derivative mechanism behind that coefficient (§8): Theorem 14 identifies it as the squared norm of a derivative-evaluation functional compressed to $\ker T_0$ and proves the resulting law, with Theorem 6 recovered as the degenerate $\theta = 0$ case (**T1, function field only**); and, kept separate from it, the ζ -side overlap model of §8.2 (**T2, numerical**) together with the parity negative control of §8.3 (**T2**). The validation gaps of this section are listed at §8.5.

Everything else is foundation or confirmation; §9 says exactly whose.

2 Foundation (prior art, stated as such)

Proposition 1 (Hallouin–Perret [8, Prop. 5, §1.2]). *[T1, cited] The Gram matrix of the normalised iterated-Frobenius classes is the Toeplitz matrix (1).*

Theorem 2 (Hallouin–Perret [8, Thm. 6, §1.3]). *[T1, cited] Let d be the rank of the Frobenius space F . Then d is the minimal integer such that $\text{Gram}(\gamma^0, \dots, \gamma^d)$ is singular, a generator of that kernel exists over $\mathbb{Q}$, and the resulting Frobenius eigenvalues are algebraic integers of modulus $\sqrt{q}$.*

Theorem 3 (Hallouin–Perret [8, Thm. 36(ii), App. A.2]). *[T1, cited] If a real symmetric positive semi-definite Toeplitz matrix of size $n + 1$ has rank n and kernel vector $(a_0, \dots, a_n)$, then $P = \sum a_j X^j$ has n distinct roots, all of modulus 1.*

Theorem 3 is step (i) of the pipeline; it is the Carathéodory–Fejér corollary that CvS [5, Cor. 1.1] prove again C^* -algebraically. Their Lemma 33 (rank = largest non-zero *leading* minor) and Lemma 34 (the $(\gamma_i \pm \gamma_{n-i})/\sqrt{2}$ parity split) are the computational tools we use throughout. CvS themselves point to [8] for the function-field connection and write that “the key difficulty in this context, then, becomes the verification that zero is indeed the (simple) minimal eigenvalue of T ” [5, Introduction].

3 The control experiment

3.1 Genuine spectra

[T1 for the arithmetic, T2 for the spectra; `src/stage1.py`, `data/stage1_spectra.csv`] Four curves ($y^2 = x^3 + x/\mathbb{F}_5$, $y^2 = x^5 + x^2 + 1/\mathbb{F}_3$, $y^2 = x^7 + x + 1/\mathbb{F}_5$, $y^2 = x^7 + x^3 + x + 1/\mathbb{F}_3$; $g = 1, 2, 3$) with brute-force point counts, L -polynomials by Newton’s identities plus the functional equation, and exact integer power sums. All 40 rows have $\lambda_{\min} \geq 0$; $\lambda_{\min}(R)$ decreases monotonically (Cauchy interlacing) and vanishes exactly at $R = d$, where the kernel polynomial’s roots are the $e^{i\theta_j}$ — Theorem 2 in action.

3.2 Planted spectra

[T1, exact over $\mathbb{Q}$; `src/stage2.py`, `src/exact_inertia.py`] We replace the Frobenius spectrum by a functional-equation-closed multiset with $|\beta| \neq 1$. Choosing rational ρ and $\cos \varphi$ makes every $t(n)$ rational, so the inertia is certified *exactly* by the Sylvester–Jacobi leading-minor rule.

configuration	planted	first R with $\lambda_{\min} < 0$	step (i) holds?	$\lambda_{\min}$ simple?
on-line control	$ \beta = 1$	never (PSD, proved)	always	fails for every $R > d$
real pair	$\rho = 1.3$	$R = 1$	always	never fails
quartet	$\rho = 1.3, \varphi = 0.7$	$R = 2$	always	never fails
two quartets	$\rho = 1.3$	$R = 4$	always	never fails
mixed	quartet + pair	$R = 4$	always	never fails

Observation 4. [T1 reasoning, T2 evidence] Step (i) holds at every applicable row of every planted configuration, including rows where $\lambda_{\min} = -642.8$: measured $\max||z| - 1| \leq 5.1 \times 10^{-48}$. Steps (i) and (iii) are therefore free of arithmetic content. This is not a discovery — CvS state Theorem 3 for an arbitrary real even distribution, so no arithmetic can enter step (i) — but the planted control makes it concrete, with numbers attached.

3.3 Degeneracy beyond the window degree

[T1 corollary of Thm. 3(i); `src/function_level.py`] Since the shift has P as both minimal and characteristic polynomial, $\ker T_R$ is the ideal generated by P truncated at degree R :

$$\ker T_R = \{P \cdot q : \deg q \leq R - d\}, \quad \dim \ker T_R = \max(0, R + 1 - d), \quad (2)$$

matched in every computed row. Hence $\lambda_{\min} = 0$ is simple exactly at $R = d$ and never after, so the CvS simple-ground-state hypothesis fails for every $R > d$; and since P is self-reciprocal, the kernel splits into even and odd parts of dimensions $\lceil \frac{m+1}{2} \rceil, \lfloor \frac{m+1}{2} \rfloor$ with $m = R - d$, both non-zero for $m \geq 1$. CvS state this in words in their Remark 2.3; we record the dimensions.

3.4 Step (iv) at the function level

[T1; `src/function_level.py`]

Proposition 5. *If $\lambda_{\min}$ is simple for infinitely many R and the normalised ground-state polynomials converge locally uniformly to $F \not\equiv 0$, then every zero of F lies on $|z| = 1$ by Theorem 3 and Hurwitz's theorem. Hence they cannot converge to a target with a zero off the circle.*

The free witness is exactly the obstruction. Measured: zero *angles* converge to $\sim 10^{-3}$ even for off-line spectra, while the coefficient distance to the target is exactly $\sqrt{2}$ (orthogonal) and the normalised $|\hat{c}_R(\beta)|$ *rises* to 0.58–0.71 out to $R = 36$ instead of falling. This makes concrete the Hurwitz step that is CvS's own Step 5.

4 The δ^2 – L^3 law

4.1 Function field

Theorem 6. [T1; proof in `notes/proofs.md`, checked by `src/proofs_check.py`] *Let the normalised spectrum be a single off-circle real pair $\{\rho, \rho^{-1}\}$ and put $a = \log \rho$. With $u_i = \rho^i$, $w_i = \rho^{-i}$ ($0 \leq i \leq R$),*

$$T_R = uw^\top + wu^\top, \quad \lambda_{\min}(T_R) = (R + 1) - \|u\| \|w\|,$$

all other eigenvalues being 0. Consequently $\lambda_{\min} < 0$ for every $R \geq 1$ as soon as $\rho \neq 1$, by strict Cauchy–Schwarz; $\lambda_{\min}$ is an even function of a ; and

$$\lambda_{\min}(T_R) = -2a^2 \sum_{i=0}^R \left(i - \frac{R}{2}\right)^2 + O(a^4) = -\binom{R+2}{3} a^2 + O(a^4).$$

For $R = 1$ this is exact: $\lambda_{\min} = -4 \sinh^2(a/2)$.

The coefficient is twice the *centred* second moment of the window; the centring falls out of the optimisation, not from a convention. Checks: the factorisation to 6.2×10^{-61} , the eigenvalues to 1.0×10^{-59} , the moment identity exactly for $R = 1, \dots, 199$, and $-\lambda_{\min}/a^2 = \binom{R+2}{3}$ to twelve printed digits for $R = 1, 2, 4, 8, 16, 32$.

4.2 Zeta side

Throughout this subsection f is real, **even**, supported in $[-L, L]$ with $\|f\|_2 = 1$, and $F = \hat{f}$; all statements here are for the even sector.

The baseline, stated explicitly. An off-line zero forces a quartet — four zeros — whereas a *simple* on-line zero at γ contributes a pair. The zeros of ζ are expected to be simple, and every computed one is, so a count-preserving comparison must be made against a *hypothetical double* on-line zero. Writing Q_{true} for the actual Weil form, the baseline below is

$$Q_0 = Q_{\text{true}} + 2F(\gamma)^2. \quad (3)$$

Two consequences, both used. First $Q_0 \geq Q_{\text{true}}$, so Theorem 7 is unaffected: a lower bound certified for Q_{true} is also one for Q_0 . Second, Theorem 8’s hypothesis $\tilde{Q} := Q_0 - 4F(\gamma)^2 \geq 0$ reads $Q_{\text{true}} - 2F(\gamma)^2 \geq 0$, i.e. exactly “every zero *other* than the one being moved lies on the critical line”. Of the three conventions this is the only one with $Q_\delta \rightarrow Q_0$ as $\delta \rightarrow 0$, and (3) is what that continuity costs: it is bought by shifting the baseline, not by comparing against ζ itself.

Moving the double zero to $\frac{1}{2} + \delta \pm i\gamma$ changes the form by

$$Q_\delta(f) - Q_0(f) = -4\delta^2(F'(\gamma)^2 + F(\gamma)F''(\gamma)) + R_4. \quad (4)$$

The two baselines, the budget, and what λ^* denotes. Call f *admissible* if it is real, even, $\text{supp } f \subset [-L, L]$ and $\|f\|_2 = 1$. Write Q_ζ for the *actual* ζ Weil form on the window, so that $Q_0 = Q_\zeta + 2F(\gamma)^2$ by (3), and set

$$b_0(L, \gamma) = \inf_{f \text{ admissible}} Q_0(f), \quad \lambda^*(L) \leq \inf_{f \text{ admissible}} Q_\zeta(f),$$

so that b_0 is the infimum of the *augmented counterfactual* baseline and $\lambda^*(L)$ denotes any *lower bound* for the *actual* unperturbed form. These are two different objects and are nowhere identified. Both are infima of a quadratic form on the stated domain; nothing here assumes the Weil form is a bounded operator on L^2 . Put

$$B_L(\delta) = c(L)\delta^2 + \frac{16}{3}L^5\delta^4e^{2L\delta}, \quad c(L) = 8L^3\left(\frac{1}{3} + \frac{1}{\sqrt{5}}\right) \approx 6.244L^3,$$

the perturbation budget: $4|F'(\gamma)^2 + F(\gamma)F''(\gamma)| \leq c(L)$ by Cauchy–Schwarz and $|R_4| \leq \frac{16}{3}L^5\delta^4e^{2L\delta}$, both proved in §B.2–B.3 of `notes/proofs.md`.

A lower bound for Q_ζ on a larger test space is *a fortiori* a lower bound on the even sector, because an infimum over a larger space cannot exceed one over a subspace; so a certificate proved on the full admissible space may be quoted here unchanged. Zhu’s certificate is used only with its documented normalisation $Q(f)/\|f\|^2$ — the one used throughout — and only on its documented range.

Theorem 7 (relative perturbation estimate; even sector). *[T1] For every admissible f and every $\delta \geq 0$,*

$$Q_\delta(f) \geq Q_0(f) - B_L(\delta),$$

and hence, taking infima over the normalised admissible sector,

$$\inf Q_\delta \geq b_0(L, \gamma) - B_L(\delta) \quad \text{and} \quad \lambda_{\min}(Q_\delta) \geq \lambda^*(L) - B_L(\delta),$$

the second because $Q_0 \geq Q_\zeta$, so any lower bound $\lambda^(L)$ for the actual unperturbed form is also one for Q_0 .*

The estimate holds *relative* to the unperturbed lower bound $\lambda^*(L)$. Replacing this baseline by zero requires independently established nonnegativity. We use the cited positivity certificate for $L \leq 0.8$: Zhu [11] certifies $\lambda^*(0.8) \geq 8.9 \times 10^{-18}$ and λ^* is non-increasing in L , so

$$\lambda_{\min}(Q_\delta) \geq 8.9 \times 10^{-18} - B_L(\delta) \quad (L \leq 0.8).$$

No such replacement is asserted here for larger windows.

Validity versus usefulness of the budget. The remainder bound inside $B_L(\delta)$ is a convergent series tail, not a truncated asymptotic expansion, so $B_L(\delta)$ is a valid budget at every δ on its proved domain. What degrades as $L\delta$ grows is its *informativeness*: the δ^4 term overtakes $c(L)\delta^2$, after which $B_L(\delta)$ no longer reflects the quadratic law. Validity and usefulness are separate questions and only the second fails there.

Theorem 8 (sharper, conditional; even sector). [T1] Define the remaining-zero form

$$Q_{\text{rest}}(f) = Q_0(f) - 4F(\gamma)^2 = Q_\zeta(f) - 2F(\gamma)^2,$$

written $\tilde{Q}$ below. Suppose every nontrivial zero other than the one being moved lies on the critical line; where the explicit formula applies, that is a sufficient input for $\tilde{Q} \geq 0$. It is not a necessary one, and $\tilde{Q} \geq 0$ does not follow from positivity of Q_ζ , from positivity of Q_0 , or from a zero-free strip: each of those leaves room for $2F(\gamma)^2$ to exceed the margin. Completing the square, $4F^2 - 4\delta^2 FF'' = (2F - \delta^2 F'')^2 - \delta^4 F''^2$, moves the FF'' allowance into $O(\delta^4)$ and gives

$$\lambda_{\min}(Q_\delta) \geq -4K(L, \gamma)\delta^2 - \left(\frac{2}{5} + \frac{16}{3}e^{2L\delta}\right)L^5\delta^4,$$

where $K(L, \gamma)$ is the exact value of the extremal problem,

$$K(L, \gamma) = \sup_{\substack{f \text{ real, even, } \text{supp} f \subset [-L, L] \\ \|f\|_2=1}} |F'(\gamma)|^2 = \int_{-L}^L u^2 \sin^2(\gamma u) du, \quad (5)$$

attained by $f \propto u \sin(\gamma u)$, with closed form

$$K(L, \gamma) = \frac{L^3}{3} - \left[\frac{L^2 \sin 2\gamma L}{2\gamma} + \frac{L \cos 2\gamma L}{2\gamma^2} - \frac{\sin 2\gamma L}{4\gamma^3} \right].$$

Corollary 9 (γ -uniform form). [T1] Since $\sin^2 \leq 1$ gives $K(L, \gamma) \leq \int_{-L}^L u^2 du = 2L^3/3$ for every γ ,

$$\lambda_{\min}(Q_\delta) \geq -\frac{8L^3}{3}\delta^2 - O(\delta^4),$$

which is the form to quote when γ is not fixed. As $\gamma \rightarrow \infty$, $K(L, \gamma) \rightarrow L^3/3$, so the uniform constant overshoots the large-ordinate value by exactly the factor 2 coming from the average of $\sin^2$.

Remark 10. $4K(L, \gamma)$ is sharp for the derivative-evaluation step: by (5) the supremum is attained, so no smaller δ^2 coefficient is available in that step. Three claims must be kept apart. **(a)** $K(L, \gamma)$ is the exact squared norm of the derivative-evaluation functional $f \mapsto F'(\gamma)$ on the normalised admissible sector. **(b)** $f \propto u \sin(\gamma u)$ attains that norm in the stated space. **(c)** Neither implies that the minimiser of the *entire* Weil form attains the same sensitivity coefficient — attaining the derivative norm need not minimise Q_δ . So $4K$ is not claimed to be a proved sharp coefficient for the optimised minimum eigenvalue; the near-saturation measured in §4.3 is T2. The constants $8L^3/3$ and $4L^3/3$ appearing below are respectively the γ -uniform upper bound and the $\gamma \rightarrow \infty$ limit of the same quantity, and ratios reported against them should be read as such.

4.3 How close the measured constant comes

[T2, src/task_final_checks.py, data/stage3_exact_constant_K.csv]

The planted ordinate is $\gamma_1 = 14.1347$. Measured $C(L)$ against the exact constant $4K(L, \gamma_1)$ of Theorem 8, and for reference against the two constants of Corollary 9:

L	0.8	1.0	1.3	1.6	2.0
measured $C(L)$	0.3108	0.8530	2.4987	4.9218	10.083
$4K(L, \gamma_1)$	0.7419	1.3427	3.1158	5.1130	10.652
$C/4K(L, \gamma_1)$	0.419	0.635	0.802	0.963	0.947
$C/(8L^3/3)$ (<i>uniform</i>)	0.228	0.320	0.427	0.451	0.473
$C/(4L^3/3)$ ($\gamma \rightarrow \infty$)	0.455	0.640	0.853	0.901	0.945

Against the exact constant the re-optimised minimiser reaches 96.3% at $L = 1.6$.

How much the $\sin^2$ weight matters, measured. The deviation $|K(L, \gamma_1)/(L^3/3) - 1|$ is

$$17.3\%, \quad 8.68\%, \quad 0.699\%, \quad 6.37\%, \quad 6.38\%, \quad 0.136\%$$

at $L = 0.6, 0.8, 1.0, 1.3, 1.6, 2.0$. It is *not* monotone: the bracketed terms in (5) oscillate in L with period π/γ , so the correction can be large at one window and negligible at a slightly larger one. At the window where positivity is certified, $L = 0.8$, it is 8.68%; at $L = 2$ it happens to be 0.136%. This is why the γ -uniform constant should not be used as a stand-in for K at a fixed ordinate.

4.4 One law, two settings: the dictionary

[T3, checked numerically in src/task_final_checks.py]

The two laws are the same statement under an explicit change of variables. For a curve over $\mathbb{F}_q$ the explicit-formula variable is $\log$ of a prime power, so the lattice index n and the ζ variable u correspond by $u = n \log q$. This fixes everything else:

	function field	ζ
window variable	$n = 0, \dots, R$	$u \in [-L, L]$
correspondence	$u = n \log q$, hence window length $R \log q = 2L$	
off-line depth	$a = \log \rho$	$\delta = \operatorname{Re} \rho - \frac{1}{2}$, with $a = \delta \log q$
Rayleigh quotient	sum over n	integral $du = \log q \, dn$: differ by $\log q$

Substituting $R = 2L/\log q$ and $a = \delta \log q$ into Theorem 6 and dividing by $\log q$:

$$-\binom{R+2}{3} a^2 \approx -\frac{R^3}{6} a^2 = -\frac{1}{6} \left(\frac{2L}{\log q} \right)^3 (\delta \log q)^2 = -\frac{4L^3}{3} \frac{\delta^2}{\log q},$$

i.e. exactly the ζ law with the large-ordinate constant $4L^3/3$, divided by the $\log q$ of the last table row. Equivalently both read (window length)³/6 in their own units.

Which factors of 2 cancel. Two enter, in opposite directions, and they cancel:

1. *Pair versus quartet.* Theorem 6 moves a reciprocal *pair*; the ζ counterfactual moves a *quartet*, twice as many zeros — a factor 2 up.
2. *$\sin^2$ averaging.* The ζ constant uses $K(L, \gamma) \rightarrow L^3/3$, half of the $\int u^2 = 2L^3/3$ that the function-field centred moment is the discrete analogue of — a factor 2 down.

Checked on one curve. For $y^2 = x^3 + x$ over $\mathbb{F}_q$, $q = 5$, with $a = 10^{-6}$, the ratio of the exact function-field $\lambda_{\min}$ to the ζ -law prediction $-\frac{4}{3}L^3\delta^2/\log q$ is

$$1.406, \quad 1.195, \quad 1.096, \quad 1.047, \quad 1.024 \quad \text{at } R = 8, 16, 32, 64, 128,$$

converging to 1 like $1 + 3/R$ — which is precisely $\binom{R+2}{3}/\frac{R^3}{6} = (1 + \frac{1}{R})(1 + \frac{2}{R})$, the only discrepancy being the $O(R^2)$ term dropped in passing from the binomial to $R^3/6$. The dictionary is therefore exact at leading order, with a fully accounted correction.

5 The ζ side as a control

[T2; src/stage3_converge.py, src/stage3_connes.py]

We assemble the window form from Zhu’s geometric side in three closed-form bases (two even, one odd), moving everything to u -space by Parseval so no oscillatory quadrature is needed, and summing the archimedean m -series analytically via digamma, Lerch Φ and Hurwitz ζ . The odd-sector pole term $-2P_j P_k$ — the sign flips because $F(-i/2) = -F(i/2)$ for odd f — is derived here.

Gate A. Under RH the explicit formula gives $Q(f) = \sum_{\rho} |F(\gamma_{\rho})|^2$. The assembled geometric side matches the zeros side to 1.63×10^{-7} (even, vanishing at $\pm L$), 2.26×10^{-7} (odd) and 4.23×10^{-3} (full cosine, which jumps at $\pm L$) at $K = 800$ zeros, each decaying at the rate that basis’s smoothness predicts (γ_K^{-3} versus γ_K^{-1}). *Control:* flipping the pole sign inflates the discrepancy to 5.35 and 0.212, so the test discriminates.

Gate B. $\lambda_{\min}(N)$ decreases monotonically in two independent complete even bases to 2.2702×10^{-17} and 2.3357×10^{-17} at $N = 24$, landing on Zhu’s certified $[8.9 \times 10^{-18}, 2.27 \times 10^{-17}]$. Zhu normalises by $Q(f)/\|f\|^2$, the same normalisation used here. We report the inequality, not the matching digits: the sequence is still falling.

Connes’s window. At $2L = \log 13$, $N = 36$, 140 digits: $\lambda_{\min} = 8.977 \times 10^{-52}$, simple and even, with the ground state’s Fourier zeros matching γ_1 to 7.2×10^{-48} and tracking roughly $N/2$ leading zeros. This reproduces at smaller scale what Groskin [6] computes at $N = 100$ –250.

Simplicity and evenness. [T2, $N = 12$, 40 digits] At $L \in \{0.5, 0.8, 1.0, \frac{1}{2} \log 13\}$ the ground state is simple with relative gap 2×10^4 to 4×10^5 and *even* in every case, the odd sector sitting 2–5 orders above. The $L = 0.8$ entry is consistent with Zhu’s certificate; the other three carry no external certificate and are T2 only.

6 Detection onset

[T2; src/stage3e_certify.py, data/stage3e_certified_onset.csv]

Precedent. Bombieri [1, §13] already ran this experiment in 2000: the first $N \leq 160$ zeros of ζ together with “a fictitious zero ρ_0 off the critical line, together with their images by complex conjugation and reflection” — the same quartet — at $\rho_0 = 0.52 + 3.14i$, on windows $E = [-t, t]$, and found a critical value t_c beyond which the negative eigenvalue stays bounded away from 0. He also separates even and odd eigenfunctions. Our contribution is to resolve that critical window in δ and in basis size, and to identify it with Zhu’s bandwidth $T^*(L) = 2\pi e^{2L}$.

The certification asymmetry. A negative Rayleigh quotient from *any* trial function certifies indefiniteness. A finite basis that finds none certifies nothing. We therefore grow the basis (low block $\cup$ resonant block $\{k : w_k \approx \gamma\}$) and report, for each planted ordinate, the smallest window at which *some* trial function goes negative. The geometric side supplies the on-line part exactly; truncating the zeros sum omits positive mass $\sim 10^{-6}$ against a signal $\sim 10^{-27}$ and cannot certify anything near the onset.

planted at	γ_n	$L_{\text{pred}} = \frac{1}{2} \log \frac{\gamma}{2\pi}$	onset, small basis	onset, largest basis
γ_1	14.13	0.405	$1.3\times$	$1.3\times$
γ_5	32.94	0.828	$1.0\times$	$1.0\times$
γ_{10}	49.77	1.035	not found	$1.0\times$
γ_{30}	101.3	1.390	not found	$0.85\times$
γ_{80}	201.3	1.733	not found	$1.15\times$

The onset tracks L_{pred} within about $\pm 30\%$ and moves *earlier* as the basis grows. Rows whose magnitude falls below the measured noise floor (the geometric control is PSD by construction, so a negative value there *is* the floor) are excluded and flagged in the data file.

What detection actually needs — and what it does not. Two conditions are visible in the data: the window must reach the ordinate, $L \gtrsim L_{\text{pred}}(\gamma_n)$, and the basis must *contain modes resonant with γ_n* , i.e. indices near $k^* \approx \gamma_n L / \pi$ — which is what the resonant block supplies. It does *not* require a basis large enough to annihilate the first n zeros. The dimension needed is far below the top mode index: γ_{30} is detected with dimension 46 and top index 45, and γ_{80} with dimension 59 and top index ≈ 134 , whereas annihilating the first n zeros in a contiguous basis would need $N \approx 2n$, i.e. 60 and 160. The $N/2$ rule of §5 governs a different question — how many leading zeros a *contiguous* basis can annihilate — and is not the detection criterion. Detection only needs some f in the subspace with $Q_{\text{geom}}(f) + 2F(\gamma)^2 < 4\delta^2(F'^2 + FF'')$, which the sparse “low block $\cup$ resonant block” subspace achieves at a small fraction of the dimension a contiguous basis would need.

Weak δ -dependence, explained. Detection is $C(L)\delta^2 > \lambda^*(L)$. By Zhu’s Landau–Widom law $\lambda^*(L)$ falls super-exponentially in L while $C(L) \sim L^3$ grows polynomially, so a factor in δ^2 shifts the crossing only logarithmically. Measured: a tenfold change in δ moves the onset from $L = 0.5$ to $L = 0.6$.

7 The recorder split

[T2; src/task2_inertia_reconcile.py]

Write $T_R = A - 2M$. In the function field the pole part of A is $q^{|n|/2} + q^{-|n|/2} = u_i w_j + w_i u_j$, an exact rank-2 *hyperbolic* plane whose scale grows like $q^{R/2}$ against a fixed $2g \cdot I$; so A is eventually indefinite under both conventions tested, M is indefinite, and $S = T_R$ is PSD. This is structural.

On ζ windows it is not. With $A = \text{Pole} + \text{Arch} - \log \pi \cdot I$, $2M = \text{Prime}$ one finds A with negative index 1 for $L \leq 0.8$ and 2 at $L = \frac{1}{2} \log 13$, identical in all three bases. But grouping $-\log \pi$ with the primes instead makes **both A and M positive definite** at $L \leq 0.8$. The reason is structural: for even f the ζ pole term is $h(i/2) + h(-i/2) = 2F(i/2)^2 \geq 0$, rank one and *definite*, not hyperbolic. There is no hyperbolic plane on the ζ side to find. We therefore state the convention ($A = \text{Pole} + \text{Arch} - \log \pi \cdot I$) and claim Block F only for the function field.

8 Projected-derivative sensitivity and near-null overlap

Theorem 6 gives the function-field δ^2 law for one real reciprocal pair, with the coefficient $\binom{R+2}{3}$ appearing as a centred second moment of the window. This section identifies the mechanism behind that coefficient — it is the squared norm of a *derivative-evaluation functional compressed to the kernel of the unperturbed form* — proves the resulting law for a family of deformations containing Theorem 6's as its degenerate member, and then reports, as a separate and strictly numerical matter, how much of the measured ζ -side sensitivity the same overlap quantity accounts for.

Three things are kept apart throughout and are never combined into a single claim: a **proved** function-field statement (§8.1), a **numerical** model on the ζ side (§8.2), and a **negative control** (§8.3). Nothing here bears on the Riemann Hypothesis, supplies a positivity certificate, or describes dynamics that move zeros of ζ .

8.1 The function-field projection law

[T1; proofs in notes/projection_sensitivity.md §§B–D', checked by extensions/projection-formula/src/stage1_ff.py, src/stage1_regression.py, src/stage1_rank.py]

Let $C/\mathbb{F}_q$ be an honest hyperelliptic curve of genus g with normalised inverse Frobenius roots $e^{\pm i\theta_j}$, $\theta_j \in (0, \pi)$ distinct, and let $t(n) = p_n q^{-n/2} = \sum_j 2 \cos(n\theta_j)$, $T_R[i, k] = t(|i - k|)$, $P_c(z) = \sum_{i=0}^R c_i z^i$.

Proposition 11 (Gram identity). [T1] For $c \in \mathbb{R}^{R+1}$, $c^\top T_R c = \sum_{j=1}^g 2|P_c(e^{i\theta_j})|^2$; for $c \in \mathbb{C}^{R+1}$, $c^* T_R c = \sum_{j=1}^g (|P_c(e^{i\theta_j})|^2 + |P_c(e^{-i\theta_j})|^2)$. Consequently $E := \ker T_R = \{c : P_c(e^{i\theta_j}) = 0 \ \forall j\}$, of dimension $R + 1 - 2g$ for $R \geq 2g$.

The coefficient 2 is the number of normalised roots in a conjugate pair, not a normalisation: the unindexed identity sums $|P_c(\hat{\alpha})|^2$ over all $2g$ roots $\hat{\alpha}$, and for real c the two members of each pair contribute equally. For complex c they do not, which is why the second form carries no factor 2 and the conjugate direction must be retained. This is the classical zero-side Gram identity, read through point-count data; it is the reason $T_R \succeq 0$ for free, and it transfers no positivity to the ζ side.

For a single index $\theta := \theta_1$ and $a \in \mathbb{R}$ put

$$t_a(n) := t(n) + 2 \cos(n\theta) (\cosh(na) - 1), \quad T_a := \text{Toeplitz}(t_a). \quad (6)$$

Remark 12 (a deformed spectrum need not come from a curve). [T1] No reciprocal pair realises (6): $\{e^{a+i\theta}, e^{-a-i\theta}\}$ has power sums $2 \cosh(na + in\theta)$, which are non-real for $a \neq 0$. The conjugation- and reciprocal-closed quadruple $\{e^{\pm a \pm i\theta}\}$ has real power sums $4 \cos(n\theta) \cosh(na)$, whose increment over the *doubled* angle is exactly twice that of (6). So (6) is the symmetrised deformation: a legitimate real symmetric Toeplitz family, but *not* the point-count sequence of a curve for $a \neq 0$. Every statement below concerns the matrix family, not a family of curves. For the genuine quadruple the coefficient in (9) doubles.

Since $\cosh$ is even, $t_a(n) - t(n) = a^2 n^2 \cos(n\theta) + a^4 n^4 \cos(n\theta)/12 + O(a^6)$ converges for every a , so $a \mapsto T_a$ is an entire family, *even* in a :

$$T_a = T_0 + a^2 B + a^4 C + O(a^6), \quad B[i, k] = (i - k)^2 \cos((i - k)\theta). \quad (7)$$

Only after this evenness is established is one entitled to write $T_\varepsilon = T_0 + \varepsilon^2 B + O(\varepsilon^4)$: the absence of a cubic term is a consequence, not a hypothesis.

Write $S_m(c) = \sum_{i=0}^R i^m c_i e^{ii\theta}$, so $S_0(c) = P_c(e^{i\theta})$. Expanding $(i - k)^2$ gives $c^\top B c = 2 \operatorname{Re}(S_2(c) \overline{S_0(c)}) - 2|S_1(c)|^2$, and on E , where $S_0 = 0$,

$$(P_E B P_E)|_E = -2 P_E (u u^\top + v v^\top) P_E|_E, \quad u_i = i \cos(i\theta), \quad v_i = i \sin(i\theta). \quad (8)$$

The coefficient is exactly -2 ; the domain is $E \subset \mathbb{R}^{R+1}$ of dimension $R + 1 - 2g$; and $\|(P_E B P_E)|_E\| \leq 2(\|u\|^2 + \|v\|^2) = 2 \sum_{i=0}^R i^2 = R(R+1)(2R+1)/3$.

Remark 13 (the compression has rank two, not one). [T1] $|S_1(c)|^2 = \langle c, u \rangle^2 + \langle c, v \rangle^2$ is a sum of *two* real rank-one forms on the real coefficient space. A modulus square is not automatically one real rank-one operator: on a real coefficient space the real and imaginary directions must both be retained unless a symmetry genuinely removes one, which here happens exactly when $\theta \in \{0, \pi\}$. Numerically the rank of (8) is exactly 2 at every tested (curve, R) with $\theta \in (0, \pi)$ and exactly 1 at $\theta = 0$. Note also that $S_1(c) = z P'_c(z)|_{z=e^{i\theta}}$ and $\frac{d}{d\theta} P_c(e^{i\theta}) = i S_1(c)$: the *moduli* agree with $|P'_c(e^{i\theta})|$, but $S_1(c)$ is not $P'_c(e^{i\theta})$, and the unimodular factor matters precisely because it rotates u and v into each other.

Theorem 14 (projected-derivative sensitivity). [T1] *Let T_0 be any real symmetric positive semi-definite Toeplitz matrix on $\mathbb{R}^{R+1}$, and let θ be the angle deformed in (6). Assume (H1) $E := \ker T_0 \neq \{0\}$; (H2) $P_c(e^{i\theta}) = 0$ for every $c \in E$, i.e. the deformed angle carries an atom of the representing measure of T_0 ; (H3) a positive complementary gap $\Delta_0 := \min\{\lambda \in \text{spec}(T_0) : \lambda > 0\} > 0$, automatic in finite dimensions unless $T_0 = 0$, but recorded because the remainder depends on it quantitatively; and (H4) analyticity of $a \mapsto T_a$, established in (7). Then*

$$\lambda_{\min}(T_a) = a^2 \lambda_{\min}((P_E B P_E)|_E) + O(a^4) = -2a^2 \max_{c \in E, \|c\|=1} |S_1(c)|^2 + O(a^4) \quad (a \rightarrow 0). \quad (9)$$

Two families satisfy (H1)–(H4), and both are used below. The **main case** is an honest curve with $R \geq 2g$ and $\theta = \theta_1$: Proposition 11 gives $\dim E = R + 1 - 2g \geq 1$ and $P_c(e^{i\theta_j}) = 0$ on E for every j , so (H1) and (H2) hold, and T_0 has rank $2g$. The **degenerate case** is $\theta = 0$ with $t_0(n)$ constant, where T_0 is a positive multiple of $\mathbf{1}\mathbf{1}^\top$, E is the mean-zero hyperplane and $P_c(1) = \sum_i c_i = 0$ on it; this is Theorem 6’s family, and stating the hypotheses this way puts it *inside* the theorem rather than alongside it. The $O(a^4)$ constant is governed by the quartic diagonal term $P_E C P_E$ and by the second-order coupling to the complement, of size $\|P_E B P_{E^\perp}\|^2 / \Delta_0$. It therefore degrades as the complementary gap Δ_0 shrinks and as the window degree grows, since $\|B\| = O(R^2)$: (9) is an asymptotic law in a at fixed (R, θ, C) and is **not** uniform in R .

Two degenerate cases are stated separately. If $R < 2g$ then $E = \{0\}$, $T_0 \succ 0$, and (9) does not apply; instead $\lambda_{\min}(T_a) = \lambda_{\min}(T_0) + O(a^2)$ is an ordinary first-order shift of a strictly positive eigenvalue, with no projection interpretation. If $\max_{c \in E} |S_1(c)|^2 = 0$, then (9) says only $\lambda_{\min}(T_a) = O(a^4)$; **no nonzero fourth- or sixth-order coefficient may be inferred from the vanishing of the quadratic one**, since the next-order combination (a Schur-complement term together with $P_E C P_E$) can itself vanish. Neither case was encountered in any tested cell, and no claim is made about them.

Regression against Theorem 6. Theorem 6’s family is exactly the $\theta = 0$ member of (6): there $t_0(n) \equiv 2$, $T_0 = 2\mathbf{1}\mathbf{1}^\top$ has rank one, $E = \{c : \sum_i c_i = 0\}$, $v = 0$ and $u_i = i$. Since P_E projects off $\mathbf{1}$, $(P_E u)_i = i - R/2$ and

$$\max_{c \in E, \|c\|=1} |S_1(c)|^2 = \|P_E u\|^2 = \sum_{i=0}^R \left(i - \frac{R}{2}\right)^2 = \frac{R(R+1)(R+2)}{12},$$

so (9) returns $\lambda_{\min}(T_a) = -R(R+1)(R+2)a^2/6 + O(a^4) = -\binom{R+2}{3}a^2 + O(a^4)$, which is Theorem 6’s coefficient, and in the $\varepsilon = \rho - 1$ normalisation of `notes/proofs.md` Remark A6 reads $-R(R+1)(R+2)\varepsilon^2/6 + O(\varepsilon^3)$. The two derivations are independent — Theorem 6 diagonalises the rank-two matrix $uw^\top + wu^\top$ directly — so the agreement identifies the paper’s *centred second moment of the window* with the squared norm $\|P_E u\|^2$ of the projected derivative-evaluation functional. Verified exactly ($< 10^{-50}$, 60 digits) at $R = 1, 2, 4, 8, 16, 32, 64$. For $\theta \neq 0$ the coefficient is *not* $\binom{R+2}{3}$: it is $-2 \max_E |S_1|^2$, depending on θ and on E , and **no universal cubic scaling in R is claimed**.

What “the error scales like ε^2 ” means. The preregistered criterion (`extensions/projection-formula/PREREGIST` K1, committed alone and before any computation) names the *relative* error of the predicted eigenvalue, $\text{rel}(a) = |\lambda_{\min}(T_a) - \text{pred}(a)|/|\text{pred}(a)|$ with $\text{pred}(a) = -2a^2 \max_E |S_1|^2$. Since $\text{pred} = \Theta(a^2)$ while the residual is $\Theta(a^4)$, $\text{rel}(a) = \Theta(a^2)$. It is therefore neither the raw residual nor the residual divided by a^2 , but the dimensionless relative error of the a^2 coefficient. At $R = 1$, $\theta = 0$, where `notes/proofs.md` Remark A7 gives $\lambda_{\min} = -4 \sinh^2(a/2)$ exactly, $\text{rel}(a) = a^2/12 + O(a^4)$, and the measured $\text{rel}(a)/a^2 = 0.0833\dots$ matches $1/12$ at $a = 10^{-2}, 10^{-3}, 10^{-4}$.

8.2 The ζ -side overlap model

[T2 throughout; `extensions/projection-formula/src/stage2_zeta.py`, `src/stage5_audit_norm.json`]

What follows is a **numerical model**, defined before it is compared with anything. It is not a continuum theorem, not a certified positivity statement, and not an established prolate projection law.

The basis is the quarter-wave even family $\cos((2k+1)\pi x/2L)$, $k < N$, with $(L, N, \text{dps}) = (0.8, 16, 70), (1.0, 18, 80), (1.3, 20, 90), (1.6, 22, 100), (2.0, 26, 120)$; it is orthogonal with $\int_{-L}^L f_j f_k = L \delta_{jk}$, so orthonormalisation is exact division by $\sqrt{L}$. Three forms are kept distinct and are nowhere identified: the actual window form $Q_{\zeta, N}$; the augmented counterfactual baseline $Q_{0, N} = Q_{\zeta, N} + 2F(\gamma)^2$, the finite-dimensional instance of the baseline (3), which is what the sweep perturbs; and the remaining-zero form $Q_{0, N} - 4F(\gamma)^2$ of Theorem 8. The baseline eigenvalue entering every reported quantity is $\lambda_{\min}(Q_{0, N})$; the minimum of $Q_{\zeta, N}$ is **not** substituted for it anywhere, notwithstanding that the two agree to all eight printed digits at every L because the window minimiser nearly annihilates $F(\gamma_1)$.

For each displacement $\delta \in \{10^{-1}, \dots, 10^{-12}\}$ the band $E_{(N, \tau)} := \text{span}\{\text{eigenvectors of } Q_{0, N} \text{ with eigenvalue } \leq \tau(\delta) := 4K(L, \gamma)\delta^2\}$ is a **cutoff-defined near-null band, not a kernel**: $Q_{0, N}$ is strictly positive definite in every cell, with $\lambda_{\min}$ from 2.5×10^{-17} at $L = 0.8$ to 1.1×10^{-67} at $L = 2.0$. The cutoff rule was fixed in the preregistration and is not a fitted threshold. The measured response is the single finite-difference ratio $C(\delta) = (\lambda_{\min}(Q_{0, N}) - \lambda_{\min}(Q_{\delta, N}))/\delta^2$ at each δ independently, with no free parameter and no fitted slope.

The overlap ratio is $r_{(N, \tau)} = \|\Pi_{E_{(N, \tau)}} v\|^2 / \|\Pi_{\text{basis}} v\|^2$ with v the representer of $c \mapsto F'(\gamma; c)$. The preregistration defined the denominator as the *continuum* norm $\|v\|^2 = K(L, \gamma)$; the implementation used the *finite-basis* projection $\|\Pi_{\text{basis}} v\|^2$. The implemented definition is retained, and the omitted tail is reported rather than the denominators interchanged: $\|\Pi_{\text{basis}} v\|^2 / K$ equals 0.9343, 0.9300, 0.9872, 0.9785, 0.999996 at the five windows, i.e. omitted tails of 6.6%, 7.0%, 1.3%, 2.2%, 4×10^{-6} . Because the prediction as computed is $4K \cdot r_{(N, \tau)}$, mixing a continuum amplitude with a finite-basis denominator **inflates it by up to 7.5% at $L = 1.0$** — more than the agreement it is compared against. Both normalisations are tabulated below.

The planted displacement expands as $Q_\delta - Q_0 = \delta^2 P + O(\delta^4)$, where P is the matrix of the quadratic form $c \mapsto -4[F'(\gamma; c)^2 + F(\gamma; c)F''(\gamma; c)]$ — that is, of (4) itself, the *full* second-order perturbation with FF'' included, the same term that Theorem 7 bounds inside its budget and that Theorem 8 moves into $O(\delta^4)$ by completing the square. Two predictions are therefore distinguished, and **the FF'' term, the variation of the baseline inside the band, the coupling to the complement and the fourth-order remainder are not discarded in order to obtain a squared norm**:

- the *squared-norm heuristic* $4K(L, \gamma) r_{(N, \tau)}$, which keeps only the $vv^\top$ part of (4) and drops the rest;
- the *compressed operator* $\lambda_{\min}(\Lambda_E + \delta^2 \Pi_E P \Pi_E)$, which retains FF'' , the exact baseline eigenvalues Λ_E of the band and the within-band coupling, and omits only the coupling to $E^\perp$ and the δ^4 remainder.

The sweep has $5 \times 12 = 60$ cells. Four of them, all at $L = 0.8$ with $\delta \leq 10^{-9}$, have $\dim E_{(N,\tau)} = 0$: there the prediction is identically zero and the relative error is 1 by construction, so they are excluded. Over the remaining 56 cells the relative error $|C - \text{pred}|/|C|$ is:

prediction	median	mean	within 3%
squared norm, as implemented ($/\ \Pi_{\text{basis}}v\ ^2$)	4.06%	527.8%	25/56
squared norm, consistently normalised ($/K$)	1.94%	489.8%	35/56
compressed operator (retains FF'')	0.23%	1.5%	48/56

The agreement commonly summarised as “about 3%” is thus a property of the *compressed second-order operator*, not of the squared-norm law; quoting it for the latter overstates that law, and quoting it for the former understates it.

The two squared-norm means are three orders above their own medians, and that gap is the point: both are dominated by exactly two of the 56 cells, $(L, \delta) = (0.8, 10^{-5})$ and $(1.0, 10^{-10})$, where C has already collapsed while the cutoff still admits a two-dimensional band. These are *retained*, not excluded; dropping them would bring the two means to 7.7% and 5.3%, which is reported here only to locate the effect. The mechanism is not what fails there — the compressed prediction is accurate to 10^{-5} and 10^{-6} at those same two cells — it is the preregistered cutoff lagging the collapse of C . Medians are therefore the summary to read, and no percentage in this work is taken across a zero crossing of the prediction.

Remark 15 (what the overlap model does and does not establish). [T3] Within the tested cells the projected overlap accounts for the bulk of the measured δ^2 response, and retaining the full second-order operator improves the account by roughly an order of magnitude over the squared-norm heuristic. This is a finite, basis- and cutoff-dependent numerical observation at five windows. It is not a proof that an overlap law holds in the continuum, it does not certify positivity, and it describes a response of the quadratic form to a *planted* displacement — not any dynamics that move zeros of ζ . A δ -staircase alignment hypothesis was also preregistered and tested; on audit the test proved uninformative against its own null and **no staircase claim is made** (notes/projection_sensitivity.md §E.4).

8.3 Parity: a negative control

[T2; extensions/projection-formula/src/stage3_parity.py, src/stage3_revalidate.py]

Zhu [11] measures a systematic parity asymmetry of the window floors, with sector ratio $\lambda_1^{\text{odd}}/\lambda_1^{\text{even}} \approx 10^{2.07L+1.32}$ over $L \in [0.5, 1.2]$, and proposes that on the zeros side it “must instead reflect the constraint $F_o(0) = 0$, which denies odd test functions the zero-free frequency interval $[0, \gamma_1)$ in which even minimizers park most of their mass”, leaving “its quantitative law, and whether the odd sector obeys a shifted version” of his decay law to a separate study. We tested the stated mechanism, in two forms.

In the full even cosine basis $\int f_k = 0$ for every $k \geq 1$ and $2L$ for $k = 0$ (confirmed, $\max_{k \geq 1} |F_k(0)| \leq 2.2 \times 10^{-81}$), so imposing $F(0) = \int f = 0$ is exactly deleting the constant mode. Comparing the constrained-even floor with the odd floor at three basis sizes per window, the ratio is 54, 103, 332, 872 at $L = 0.5, 0.6, 0.8, 1.0$ at the largest basis, against a preregistered tolerance of a factor 3. The ratio is far better conditioned than the floors themselves: at $L = 1.0$ it moves only from 893 to 872 while the floors fall by three orders of magnitude towards convergence. Individual floors from the original single-basis run are *not* quoted as converged values — at $L = 1.0$, $N = 18$ the even floor sits $896\times$ above Zhu’s reference value, falling to $2.3\times$ at $N = 30$, every floor approaching from above as finite- N variational upper bounds must.

Because $F(0) = 0$ is a single linear condition and strictly weaker than oddness, the interval reading was tested directly, by measuring the fraction of $|F|^2$ mass on $[0, \gamma_1)$ for each minimiser (Plancherel normalisation, validated numerically). The odd minimiser carries 99.67% to 99.99%

of its mass there — essentially as much as the even minimiser’s 99.99%, with the constrained-even minimiser lowest at 96.4%.

Observation 16 (scoped). [T2] Within the tested setting, imposing $F(0) = 0$ on the even sector does not reproduce the odd-sector floor; the single-condition explanation is unsupported. Within the tested range $L \leq 1.0$ the mass-parking refinement is not the discriminator either, both parities concentrating essentially all their $|F|^2$ mass on $[0, \gamma_1)$ — a structural consequence of $\gamma_1 = 14.13$ exceeding the retained basis frequencies at these windows, and hence a limitation of the tested range rather than a refutation at large L .

This does **not** show that parity is irrelevant: the sectors differ by two to four orders of magnitude and follow Zhu’s geometric ratio law closely, so parity carries near-null structure that the single linear constraint does not capture, and what that structure is remains unidentified here. Nor does it address the question Zhu actually leaves open — the quantitative law of the sector ratio, and whether the odd floor obeys a shifted decay law — which is untouched.

8.4 On the prolate question

[T3]

Prior art exists for prolate spheroidal wave functions approximating the lowest eigenvector of a truncated Weil form, and it is weaker than an identification. Connes [4] §6.4 describes “a construction numerically justified of the eigenvectors associated to the first minuscule eigenvalues of QW_λ , using prolate spheroidal wave functions”, giving “an educated guess for an approximation” of the lowest eigenvector; Connes–Consani–Moscovici [3] §7 cite the same as “the observation . . . that the eigenfunction associated with the lowest eigenvalue of QW_λ is well approximated by prolate spheroidal wave functions”. It concerns the *single* lowest eigenfunction, it is explicitly numerical, and even simplicity of that eigenvalue is listed as a remaining step. It also concerns the multiplicative window $[\lambda^{-1}, \lambda]$ and the operator QW_λ , a different construction from the additive-window Galerkin form used here, and no identification between the two is established in any source we were able to check.

Accordingly the object of §8.2 is described as the projection onto the *computed near-null subspace* $E_{(N,\tau)}$ of $Q_{0,N}$, with its cutoff stated, and we do not write that the near-null space is the prolate space. The prolate quantities $1 - \chi_k$ were not computed; at the 10^{-50} scale they require dedicated prolate code, and their absence is a scope limitation. In Zhu [11] the prolate connection enters only as the rate constant $2\pi^2$ of an empirical decay law, matching the Landau–Widom eigenvalue-plunge rate, and not as any subspace identification. Several directly relevant sources were not accessible to us and are marked as unchecked in `notes/projection_sensitivity.md` §F; absence of a statement from the sources we did check is not evidence of novelty.

8.5 Validation gaps in this section

Listed explicitly, because the rest of this paper is held to a stated tier and this section should be too. None of these affects §§1–7 or Theorems 6–8.

1. **Theorem 14 is about a matrix family, not a family of curves.** For $a \neq 0$ the sequence (6) is not the normalised point-count sequence of any curve (Remark 12); and the law is asymptotic in a at fixed window degree, with an $O(a^4)$ constant that grows like R^2 and degrades as the complementary gap shrinks. It is **not uniform in R** , and the measured relative error at $\theta = 0$ grows from $0.083 a^2$ at $R = 1$ to $54.3 a^2$ at $R = 32$.
2. **The two excluded degenerate cases carry no higher-order claim.** When $E = \{0\}$ ($R < 2g$) the projection law does not apply at all; when $\max_{c \in E} |S_1(c)|^2 = 0$ the law gives only $\lambda_{\min}(T_a) = O(a^4)$, and no nonzero quartic or sextic coefficient is inferred. Neither case arose in any tested cell, so neither is checked rather than merely argued.

3. **The overlap model of §8.2 is numerical and narrow:** five windows, one basis, $N \leq 26$, and a near-null band defined by a cutoff rather than an exact kernel. Nothing is proved about the continuum limit, about $N \rightarrow \infty$, or about basis-independence.
4. **The overlap denominator deviates from the preregistration.** The preregistered ratio divides by the continuum norm $K(L, \gamma)$; the implementation divides by the finite-basis projection norm. The implemented definition is retained and the omitted tail reported, but the consequence is that the prediction as computed is **inflated by up to 7.5%** at $L = 1.0$ — larger than the agreement being reported. Both normalisations are tabulated; neither is presented as the other.
5. **Four of the 60 cells are excluded from the agreement table:** those with $\dim E_{(N, \tau)} = 0$, all at $L = 0.8$ with $\delta \leq 10^{-9}$, where the prediction is identically zero and the relative error is 1 by construction. The two cutoff-lag outliers at $(L, \delta) = (0.8, 10^{-5})$ and $(1.0, 10^{-10})$ are *not* excluded — they are among the 56 cells tabulated, and they are what carries the squared-norm means to 527.8% and 489.8% against medians of 4.06% and 1.94%. Medians are the summary to read; a mean here is not a robust statistic.
6. **The preregistered δ -staircase hypothesis is withdrawn, not confirmed.** Its acceptance test as implemented was passed by 89% of all available brackets, so the reported 11-of-12 carries no information; under the preregistered criterion the result falls below its own null. No staircase claim is made here, and the hypothesis is neither supported nor refuted — the test was simply uninformative.
7. **The parity floors are finite- N variational upper bounds, not converged values.** At $L = 1.0$ the original single-basis run sat $896\times$ above Zhu’s reference value, falling to $2.3\times$ at $N = 30$. The *ratio* is far better conditioned than the floors and the negative verdict is stable in N , but no individual floor here should be quoted as converged.
8. **The mass test cannot discriminate at the windows tested.** For $L \leq 1.0$ the interval $[0, \gamma_1)$ contains nearly all the $|F|^2$ mass of *both* parities, so its failure to separate them is a limitation of the tested range rather than evidence against the mechanism at large L . Extending it would need L large enough that basis frequencies reach γ_1 .
9. **Four directly relevant prior-art sources were not accessible** and are unchecked: Connes–Consani, *Spectral triples and ζ -cycles*, Enseign. Math. **69** (2023) 93–148 — which is the actual origin of the prolate-approximation observation — Connes–Moscovici, PNAS **119** (2022) e2123174119; Connes–Consani, Selecta Math. **27** (2021) art. 77; and *Riemann Zeros via Weil Forms: From Prolate Functions...* (ref. [31] of [4]). Absence of a statement from the sources that *were* checked is not evidence of novelty.
10. **The prolate quantities $1 - \chi_k$ were not computed**, so §8.4 compares nothing numerically against the prolate literature; it only records what that literature does and does not assert.

9 Relation to prior work

Hallouin–Perret [8] is the function-field foundation: Proposition 5 is the window Toeplitz matrix, Theorem 6 the first-singular-size and kernel generator, Theorem 36(ii) step (i), Lemmas 33–34 the rank rule and parity split used here. Connes–van Suijlekom [5] supply the C^* -algebraic proof of step (i) and the Hurwitz step, cite [8] for the function-field connection, and state the simplicity difficulty themselves; their Remark 2.3 is our degeneracy observation in words. Bombieri [1] is the precedent for planted ζ experiments (§13) and proves the negative-eigenvalue count (Theorem 8); Yoshida [10] first studied the Weil hermitian form on functions supported

in $[-t, t]$. **Zhu** [11] supplies the certified $\lambda^*(0.8)$, the normalisation and the bandwidth T^* . **Groskin** [6, 7] supplies the large-scale Galerkin computations and the archimedean tail order. **Connes–Consani–Moscovici** [3], **Connes** [4], **Connes–Consani** [2] and **Suzuki** [9] are the surrounding program. A fuller account, with verbatim quotations and the verification method, is in `PRIOR_ART.md`.

10 Limitations

1. No rigorous lower bound on $\lambda^*(L)$ of our own. Theorem 7 is a *relative* estimate; it becomes an unconditional lower bound only where Zhu’s certificate reaches ($L \leq 0.8$). Replacing $\lambda^*(L)$ by 0 beyond that range would assume the window positivity at issue, and is not asserted (Retraction 18).
2. Theorem 14 concerns the real symmetric Toeplitz family (6), which for $a \neq 0$ is not the point-count sequence of a curve; it is asymptotic in a at fixed window degree and is not uniform in R .
3. The overlap model of §8.2 is numerical, at five windows, in one basis, with a cutoff-defined near-null band rather than a kernel; and the preregistered δ -staircase hypothesis was tested and withdrawn as uninformative, so no staircase claim is made.
4. The parity result of §8.3 is a negative control at four windows. It does not show that parity is irrelevant, and it does not address the quantitative law of the sector ratio, which Zhu [11] leaves open. The full list of §8’s validation gaps is §8.5.
5. Theorem 8 is conditional on all other zeros being on the line. Nonnegativity of the remaining-zero form $\tilde{Q} = Q_\zeta - 2F(\gamma)^2$ is not supplied by positivity of Q_ζ , by positivity of Q_0 , or by a zero-free strip.
6. A zero-location restriction alone does not discharge the hypothesis of Theorem 8. For the audited source version and the transfer attempted in `extensions/openai-quasi-rh-audit-2026-10-07/`, no usable estimate was obtained that replaces the remaining-zero positivity hypothesis: the restriction bounds how far a zero may sit from the critical line, not the aggregate window-weighted contribution of all such zeros, and the per-zero bound used there is uniform in the ordinate and therefore not summable as it stands. This is a limitation of that particular estimate, not a general obstruction.
7. Interval certification covers the zeros-side assembly only: `mpmath`’s interval context has no digamma and no Lerch transcendent, so the geometric side could not be interval-evaluated. What is certified instead is the $\text{LDL}^\top$ witness identity $v^\top Z v = D_j$ and a measured noise floor.
8. Onset non-detections are limited by the subspaces actually searched: the largest had dimension 59 with top mode index ≈ 134 . A non-detection bounds that search, not the method.
9. $C(L)$ is measured at finite δ and finite basis; the saturation $C/(4L^3/3) \rightarrow 1$ is observed, not proved.
10. The ζ -side numbers depend on the planted ordinate, the basis and the truncation.

11 Traps this project documents

The retraction log (Appendix A) is part of the result. The reusable lessons: **(a)** truncating the zeros sum biases the form *downward* and manufactures false detections; **(b)** moving a zero and

adding a zero are two different counterfactuals, both legitimate — Bombieri studies the second — and the sharp law needs the first; **(c)** `float64` returns spurious negative eigenvalues of size 10^{-16} where the true value is exactly 0; **(d)** interval libraries have gaps, and `mpmath`’s `iv.sincpi` is itself broken (it calls a non-existent `ctx.sinpi`); **(e)** a non-detection certifies nothing; **(f)** keeping known-good regression values was the only thing that caught three sector-rename bugs.

A Retractions

Twenty corrections were logged during this work; the full list with evidence is in `REPORT.md`. Those a reader must see:

#	Retraction
1	Frobenius radius computed on P rather than its reverse (0.8 instead of $\leq 1.3 \times 10^{-51}$).
3	A tolerance-based sign test misclassified exact zeros; replaced by an error-based test plus exact rational certificates.
4	H2 as first stated was refuted by my own adversarial pass ; restated with a distinctness hypothesis. The restatement is Theorem 2.
6	“Step (iv) is free” retracted — it had been scored on zero angles, which is the wrong object. At the function level it carries content (Proposition 5).
8–9	A 40-digit run discarded; three sector-rename bugs caught only by regression values.
11	“Low-lying zeros only” retracted as structural . The height sweep had used a truncated on-line side, and non-detections were basis-limited. Redone, $\gamma_{30} \approx 101$ is detected at $0.85 L_{\text{pred}}$. <i>Amended</i> : the added-quartet counterfactual is legitimate in its own right — it is Bombieri’s — and was not the error.
12	γ_{80} “detections” below their own noise floor discarded; later detected at $1.15 L_{\text{pred}}$ with a positive control.
15	“Block F holds on the ζ side” retracted (§7).
16	Three claimed contributions are published theorems : step (i) is [8, Thm. 36(ii)], the first-singular-size result is [8, Thm. 6], the window matrix is [8, Prop. 5]; the degeneracy observation is [5, Rem. 2.3]; the negative-index count is [1, Thm. 8]; the planted experiment is [1, §13]. An earlier abstract-level novelty check had missed all of these.
17	The $N \gtrsim 2n$ detection rule retracted (§6). It contradicted both the stated limitation and the γ_{80} table row. The γ_{80} detection used dimension 59, top index ≈ 134 , against $2n = 160$; γ_{30} used dimension 46, top index 45, against $2n = 60$. The rule had been imported from §5, where $N/2$ governs a stronger question — how many leading zeros a <i>contiguous</i> basis annihilates. The table row and the limitation were correct; only the rule was wrong.
18	The unsupported replacement of the unperturbed baseline by zero outside the certified range has been withdrawn . Theorem 7 previously added that for $L > 0.8$ it “holds with $\lambda^*(L)$ replaced by 0, using no positivity input at all”. Substituting 0 for $\lambda^*(L)$ in a lower bound requires $\lambda^*(L) \geq 0$, which is the finite-window Weil positivity at issue; at $\delta = 0$ the substituted statement asserts $\lambda_{\min}(Q_0) \geq 0$, i.e. the very input it disclaimed. The preceding relative perturbation estimate is retained, and is now stated as such. The same step was the last link of the chain in §B.4 of <code>notes/proofs.md</code> and is corrected there. No numerical value, no other theorem and no table is affected.

- 19 **The δ -staircase alignment claim of the projection experiment has been withdrawn as uninformative.** Its internal report scored “11 of 12 plateau edges aligned with an eigenvalue scale” as a pass. On audit the denominator 12 is the *observed* number of edges, not a preregistered list; the acceptance window as implemented spans a factor 90 in δ where the preregistration specified a factor 3 in δ^2 , 27 times tighter; and a null model over all 55 available decade brackets passes 89% of them, so 11/12 is indistinguishable from chance, while under the preregistered criterion the result is 8/12, *below* its own 75% null. No staircase claim is made in §8. The hypothesis is neither supported nor refuted: the test was uninformative. The measured $C(\delta)$ values are unaffected.
- 20 **A prolate prior-art attribution has been corrected.** The same internal report quoted [3] as stating that the space of eigenvectors of the k lowest eigenvalues of QW_λ “corresponds to the prolate projection $\Pi(\lambda, k)$ ”, and concluded that the near-null space *is* prior art. Checked against the primary text, that paper says the *single* lowest eigenfunction “is well approximated by” prolate spheroidal wave functions, as an observation attributed to Connes–Consani (Enseign. Math. **69** (2023) 93–148); the notation $\Pi(\lambda, k)$ does not occur in it, and [4] §6.4 describes the construction as “numerically justified” and “an educated guess for an approximation”. §8.4 states the weaker, sourced version, and no identification of the computed near-null subspace with a prolate space is claimed.
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B Reproduction

Code and data: <https://github.com/innerlightr-wq/weil-window-control>. Python 3.12, mpmath 1.3.0, numpy 1.26.4, matplotlib 3.8+; no gmpy2 (installing it speeds the high-precision runs substantially).

Result	Script	Data
§3 genuine	src/stage1.py	stage1_spectra.csv
§3 planted	src/stage2.py	stage2_spectra.csv
§3 kernel	src/function_level.py	stage2_function_level.csv
Thm 6, 7, 8	src/proofs_check.py	stage3_sharp_bound.csv
C vs Cauchy–Schwarz	src/task34_extra.py	stage3_C_vs_CS.csv
$K(L, \gamma)$, dictionary	src/task_final_checks.py	stage3_exact_constant_K.csv
Gates A, B	src/stage3_converge.py	stage3_gateA/B_*.csv
Connes window	src/stage3_connes.py	stage3b_connes_zeros_N36.csv
Detection onset	src/stage3e_certify.py	stage3e_certified_onset.csv
Recorder split	src/task2_inertia_reconcile.py	stage3d_inertia_reconciled.csv
§8, in <code>extensions/projection-formula/</code> (preregistration PREREGISTRATION.md)		
Thm 14, 44 cells	src/stage1_ff.py	data/stage1_ff.json, _dps120.json
Thm 6 regression	src/stage1_regression.py	data/stage1_regression.json
Realisability, rank	src/stage1_rank.py	data/stage1_rank.json
$R < 2g$ (no kernel)	src/stage1_belowd.py	data/stage1_belowd.json
Overlap model, δ sweep	src/stage2_zeta.py	data/stage2_zeta.json
Normalisation audit	src/stage5_audit_norm.py	data/stage5_audit_norm.json
Staircase null model	src/stage5_audit_staircase.py	data/stage5_audit_staircase.json
Parity floors, mass test	src/stage3_revalidate.py	data/stage3_revalidate.json
v9 number check	src/verify_v9_numbers.py	(prints; no data file)

Acknowledgments

AI assistance was used for algebra checking, numerical verification, adversarial review, literature organization, and manuscript drafting. For §8 it was additionally used to run the preregistered experiment in `extensions/projection-formula/`, to audit that experiment’s own reported claims against their preregistered criteria and against a null model, and to check the quoted statements of [11], [3] and [4] against those primary sources; two claims from the extension’s internal report were withdrawn on that audit. All conceptual development, research direction, interpretation, and final responsibility for the content belong to the author.

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